

▼ Importing Necessary Libraries

```
!pip install sumy
import sumy
```

```
Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
Collecting sumy
  Downloading sumy-0.11.0-py2.py3-none-any.whl (97 kB)
    |#####| 97 kB 2.9 MB/s
Collecting pycountry>=18.2.23
  Downloading pycountry-22.3.5.tar.gz (10.1 MB)
    |#####| 10.1 MB 27.1 MB/s
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing wheel metadata ... done
Requirement already satisfied: nltk>=3.0.2 in /usr/local/lib/python3.8/dist-packages (from sumy) (3.7)
Collecting breadability>=0.1.20
  Downloading breadability-0.1.20.tar.gz (32 kB)
Requirement already satisfied: requests>=2.7.0 in /usr/local/lib/python3.8/dist-packages (from sumy) (2.23.0)
Collecting docopt<0.7,>=0.6.1
  Downloading docopt-0.6.2.tar.gz (25 kB)
Requirement already satisfied: chardet in /usr/local/lib/python3.8/dist-packages (from breadability>=0.1.20->sumy) (3.0.4)
Requirement already satisfied: lxml>=2.0 in /usr/local/lib/python3.8/dist-packages (from breadability>=0.1.20->sumy) (4.9.2)
Requirement already satisfied: tqdm in /usr/local/lib/python3.8/dist-packages (from nltk>=3.0.2->sumy) (4.64.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.8/dist-packages (from nltk>=3.0.2->sumy) (1.2.0)
Requirement already satisfied: regex<2021.8.3 in /usr/local/lib/python3.8/dist-packages (from nltk>=3.0.2->sumy) (2022.6.2)
Requirement already satisfied: click in /usr/local/lib/python3.8/dist-packages (from nltk>=3.0.2->sumy) (7.1.2)
Requirement already satisfied: setuptools in /usr/local/lib/python3.8/dist-packages (from pycountry>=18.2.23->sumy) (57.4.0)
Requirement already satisfied: urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1 in /usr/local/lib/python3.8/dist-packages (from requests>=2.7.0->sumy) (2.10)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.8/dist-packages (from requests>=2.7.0->sumy) (2022.12.7)
Building wheels for collected packages: breadability, docopt, pycountry
  Building wheel for breadability (setup.py) ... done
  Created wheel for breadability: filename=breadability-0.1.20-py2.py3-none-any.whl size=21714 sha256=85740c1f6adc5f75da9caed84b358:
  Stored in directory: /root/.cache/pip/wheels/5f/0d/0c/2062d8c1758b4b1a2e42b4a63e6660d9ec2ba9463cf9ee9eab
  Building wheel for docopt (setup.py) ... done
  Created wheel for docopt: filename=docopt-0.6.2-py2.py3-none-any.whl size=13723 sha256=c895edbd470f1ad6382d2fa808f41d416908f9f28:
  Stored in directory: /root/.cache/pip/wheels/56/ea/58/ead137b087d9e326852a851351d1debf4ada529b6ac0ec4e8c
  Building wheel for pycountry (PEP 517) ... done
  Created wheel for pycountry: filename=pycountry-22.3.5-py2.py3-none-any.whl size=10681845 sha256=ec3441e51e7d7532ff58ae0b1ed9c5f:
  Stored in directory: /root/.cache/pip/wheels/e2/aa/0f/c224e473b464387170b83ca7c66947b4a7e33e8d903a679748
Successfully built breadability docopt pycountry
Installing collected packages: docopt, pycountry, breadability, sumy
Successfully installed breadability-0.1.20 docopt-0.6.2 pycountry-22.3.5 sumy-0.11.0
```

- ▼ Importing TextRank from sumy library

```
from sumy.summarizers.text_rank import TextRankSummarizer
```

▼ Downloadind Dependencies

```
import nltk
nltk.download('punkt')

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
True
```

Link [Source of text]: <https://www.mtu.edu/biomedical/departments/what-is/>

```
# Text for summarization
text="""Biomedical engineering is the application of the principles and problem-solving techniques of engineering to biology and medicine. Engineering itself is an innovative field, the origin of ideas leading to everything from automobiles to aerospace, skyscrapers to sonar. According to the United States Department of Labor, the mean salary for a biomedical engineer is $101,020 with the top ten percent of biomedical engineers earning more than $130,000 annually.
"""

# Setting up parser --> I've chosen PlainTextParser but different can be chosen depending on the type of article/text to be summarized
from sumy.nlp.tokenizers import Tokenizer
from sumy.parsers.plaintext import PlaintextParser
```

```
parser=PlaintextParser.from_string(text,Tokenizer('english'))
```

Basically the whole article is copied for summarization from the above link.

▼ Instantiating the model and creating summary

```
# Instantiate TextRankSummarizer()
summarizer = TextRankSummarizer()

# Ranking sentences on the basis of TextRank algorithm and choosing top 2 sentences for summary
summary_sentences = summarizer(parser.document,2)

# Printing the summary
summary = ""

for sentence in summary_sentences:
    summary = summary + str(sentence)

print(summary)
```

☞ and developing new diagnostic tools and medical equipment; and in government for establishing safety standards for medical devices.

▼ Summary

This is evident throughout healthcare, from diagnosis and analysis to treatment and recovery, and has entered the public conscience through the proliferation of implantable medical devices, such as pacemakers and artificial hips, to more futuristic technologies such as stem cell engineering and the 3-D printing of biological organs. There are opportunities in industry for innovating, designing, and developing new technologies; in academia furthering research and pushing the frontiers of what is medically possible as well as testing, implementing, and developing new diagnostic tools and medical equipment; and in government for establishing safety standards for medical devices.